

Physics-First Radar-to-Point-Cloud Learning: Initial Training Results and Fair Baseline Evaluation

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Abstract—We present initial results on mmDar, a system that generates lidar-quality point clouds from raw millimeter-wave radar signals. Starting from the RadarHD baseline [1], which maps preprocessed radar magnitude images to occupancy grids using a U-Net, we develop a physics-first Gaussian set prediction model that operates directly on complex-valued IQ data. Our approach applies classical FFT beamforming as a fixed preprocessing stage, encodes the resulting 2D azimuth–range representation with a convolutional encoder, and decodes scene geometry through a DETR-style set prediction head that outputs parametric Gaussians in polar coordinates. A key methodological contribution is the establishment of a fair evaluation protocol: we show that the published baseline result of 0.295 m Chamfer distance relies on test-set checkpoint selection, and that an honestly validated baseline achieves 0.406 m. Against this fair reference, our physics-first model achieves a Chamfer distance of 0.318 m and a modified Hausdorff distance of 0.230 m—improvements of 22% on both metrics—while using $5.6\times$ fewer parameters (3.1M vs. 17.5M). We also document negative results from ConvLSTM temporal fusion and analyze error decomposition across range bins.

I. INTRODUCTION AND PROJECT GOAL

Generating lidar-quality point clouds from automotive millimeter-wave (mmWave) radar is an attractive prospect: radar sensors are inexpensive, compact, and robust to adverse weather [2]. However, radar measurements suffer from limited angular resolution, multipath interference, sidelobe artifacts, and speckle noise, making the mapping from radar observations to dense scene geometry fundamentally ill-posed.

The RadarHD project [1] demonstrated that an asymmetric U-Net can map stacked radar magnitude images to high-resolution polar occupancy grids, achieving a reported Chamfer distance of 0.36 m. While effective, this approach discards radar phase information during preprocessing, collapses 60+ dB of dynamic range to 8-bit integers, and relies on a hand-crafted occupancy grid output representation that is poorly suited to geometric distance metrics.

Our project investigates whether a physics-informed pipeline can improve point cloud quality by:

- 1) Preserving raw complex-valued radar data through classical signal processing rather than lossy image conversion;
- 2) Replacing dense occupancy prediction with a set-based Gaussian output parameterization trained with Hungarian matching; and

- 3) Maintaining a fair evaluation protocol with explicit train/validation/test splits and validation-based checkpoint selection.

The central research question is: *Can a physics-informed input representation and set-based output parameterization produce more accurate point clouds than image-translation baselines under honest evaluation?*

II. DATASET DESCRIPTION

A. Sensor and Data Collection

The dataset was collected using a TI AWR1843 77 GHz FMCW radar with 2 transmit and 4 receive antennas, yielding 8 virtual elements via TDM-MIMO [1]. Each frame consists of 192 chirps across 512 ADC samples. Ground truth is provided by a co-registered lidar sensor producing dense 3D point clouds. The full dataset comprises 44 trajectories totaling approximately 42,000 paired radar–lidar frames.

B. Data Splits

A critical lesson from our experiments is that trajectory-level splitting is essential. Adjacent frames within a trajectory are highly correlated; frame-level splits produce information leakage and optimistic results. We use two split configurations:

Low-ID split (for baseline comparison): 17 training / 8 validation / 19 test trajectories. The test set contains 18,575 frames and is sealed for final evaluation only.

Mixed-ID split (for Gaussian model development): 25 training / 6 validation / 13 test trajectories, including high-ID trajectories (227–250) in all partitions to reduce domain shift.

C. Input Representations

Baseline (magnitude PNGs): The RadarHD pipeline applies range FFT, azimuth FFT, magnitude extraction, 5% thresholding, and uint8 quantization. The result is a 256×64 polar image per frame; 41 consecutive frames are stacked as input channels.

Physics-first (raw IQ): Our pipeline preserves significantly more information: (1) range FFT along 512 ADC samples, (2) Doppler FFT across 64 chirps per TX, (3) TDM-MIMO Doppler phase compensation for TX2, (4) zero-Doppler extraction and TX concatenation to form an 8-element virtual array. The output per frame is an $(8, 512)$ complex64 tensor, normalized per-frame by maximum magnitude. Lidar ground

truth is subsampled to 8,192 points via farthest point sampling (FPS) and scene-filtered to $x \in [0, 10]$ m, $|y| \leq 10$ m, $|z| \leq 0.3$ m.

III. IMPLEMENTED ALGORITHMS

A. Baseline: Asymmetric U-Net (UNet1)

The baseline follows the RadarHD architecture [1], [3]: a 4-level encoder (64–128–256–512–512 channels with Max-Pool2d) and a symmetric decoder with skip connections, followed by three additional upsample-only blocks that expand the azimuth dimension from 64 to 512 bins. The output is a single-channel sigmoid activation representing occupancy probability.

Loss: $\mathcal{L} = 0.9 \cdot \text{BCE} + 0.1 \cdot \text{Dice}$. **Training:** Adam [4], lr = 7×10^{-5} , batch size 12, 50 epochs, fp32. **Parameters:** 17.5M. **Inference:** Threshold sigmoid output $\rightarrow$ extract occupied pixels $\rightarrow$ polar-to-Cartesian conversion.

B. Temporal Extension: ConvLSTM

To investigate temporal fusion, we extended the baseline with two ConvLSTM cells [5] inserted at the bottleneck (16 $\times$ 4 spatial) and deepest skip level (32 $\times$ 8 spatial). The encoder/decoder weights are shared across frames, and only the ConvLSTM cells maintain sequential state. **Parameters:** 27.5M.

C. Physics-First Gaussian Set Model

Our proposed model (Fig. 1) has three stages:

Stage 1—Classical FFT Frontend (0 learned parameters): A fixed FFT beamformer is applied per range bin: $\text{fft}(\mathbf{x}, n = 64, \text{dim} = \text{antenna})$ followed by fftshift . This produces a (64, 512) complex spectrum per frame. We extract magnitude and log-power channels, yielding (2, 64, 512) per frame. For $T=41$ frames, the input to the encoder is (82, 64, 512).

Stage 2—Deep 2D Encoder (2.38M parameters): A residual convolutional encoder processes the azimuth–range representation: $\text{Conv2d}(82 \rightarrow 64) \rightarrow \text{ResBlock}(64) \rightarrow$ three DownBlocks with stride-2 convolution (64 $\rightarrow$ 128 $\rightarrow$ 192 $\rightarrow$ 192), reducing spatial dimensions to (8, 64). GroupNorm and GELU activations are used throughout, with Dropout2d(0.1) for regularization. The output is flattened to 512 spatial tokens of dimension 192, augmented with decomposed 2D positional encodings.

Stage 3—DETR-Style Gaussian Decoder (0.72M parameters): Inspired by DETR [6] and mmPoint [7], $K = 96$ learnable query vectors attend to the encoder output through 3 transformer decoder layers (4-head attention, $d = 128$). Each query predicts 5 parameters:

$$(\mu_r, \mu_\varphi, \sigma_r, \sigma_\perp, \text{existence}) \quad (1)$$

where $\mu_r = \sigma(\cdot) \cdot r_{\max}$, $\mu_\varphi = \tanh(\cdot) \cdot \frac{\pi}{2}$, and $\sigma_\perp$ scales with range (physics-informed: angular uncertainty grows with distance). Cartesian centers are recovered as $\mu_{xy} = (\mu_r \cos \mu_\varphi, \mu_r \sin \mu_\varphi)$.

Total parameters: 3.1M (5.6 $\times$ fewer than baseline).

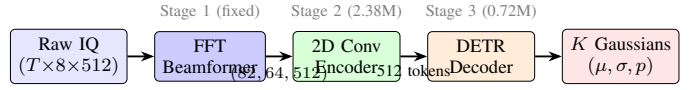


Fig. 1: Physics-first Gaussian model architecture. Stage 1 applies a fixed classical FFT beamformer; Stages 2–3 contain the only learned parameters (3.1M total).

D. Loss Function

The Gaussian model uses a composite loss with Hungarian matching [8] for one-to-one assignment between predictions and K -Means cluster centers fitted to lidar ground truth ($K = 64$ clusters per frame):

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \mathcal{L}_{\text{exist}} + 0.5 \mathcal{L}_{\text{cov}} + 0.5 \mathcal{L}_{\text{card}} + 0.1 \mathcal{L}_{\text{rep}} + 0.1 \mathcal{L}_\sigma + 0.1 \mathcal{L}_{\text{hub}} \quad (2)$$

The dominant term $\mathcal{L}_{\text{NLL}}$ is a heteroscedastic negative log-likelihood in a local radial/perpendicular coordinate frame:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{2} \left(\frac{d_r^2}{\sigma_r^2} + \frac{d_\perp^2}{\sigma_\perp^2} \right) + \log \sigma_r + \log \sigma_\perp \quad (3)$$

where $d_r, d_\perp$ are the radial and perpendicular errors between matched prediction and ground truth. The Huber range term $\mathcal{L}_{\text{hub}}$ (SmoothL1 on range with $\beta = 0.2$) was added to reduce range bias.

Training: AdamW [9], lr = 10^{-4} , cosine annealing over 50 epochs, batch size 4, gradient clipping at norm 1.0. **Augmentation:** Horizontal flip ($p = 0.5$), complex Gaussian noise at SNR 15–25 dB ($p = 0.5$), temporal frame masking of 2–6 frames ($p = 0.3$).

IV. EVALUATION PROTOCOL

A. Metrics

We report two principal distance-based metrics, both computed in Cartesian coordinates (meters) as frame-level medians across the test set:

Chamfer distance [10]: The symmetric average nearest-neighbor distance between predicted and target point sets. It measures overall coverage quality.

Modified Hausdorff distance (mod-H): $\max(\text{median}_{p \in P} \min_{q \in Q} d(p, q), \text{median}_{q \in Q} \min_{p \in P} d(p, q))$. This metric is stricter than Chamfer—it penalizes imprecise outliers and structural errors.

We also report IoU, F1, and precision/recall at 0.1 m, though these are less informative for this task due to extreme label sparsity (IoU ≈ 0.03 –0.05).

B. Checkpoint Selection Bias

A critical finding is that the published baseline result of Chamfer 0.295 m was obtained by selecting the best checkpoint on the *test set*—effectively a form of test-set overfitting. Under honest validation-based selection (epoch 35 by validation mod-H), the same architecture achieves Chamfer 0.406 m and mod-H 0.296 m, a substantial degradation (38% worse Chamfer). Table I quantifies this effect, and Fig. 4 visualizes

TABLE I: Checkpoint selection bias in the baseline U-Net (same model, same test set of 18,575 frames).

Selection Method	Chamfer (m)↓	mod-H (m)↓
Test-set selected (epoch 10)	0.295	0.189
Validation-selected (epoch 35)	0.406	0.296
Published RadarHD [1]	0.36	0.24

TABLE II: Model comparison under fair evaluation. All results are frame-median on the 18,575-sample test set except [†] which uses a 13-trajectory mixed-ID test set.

Model	Params	Chamfer↓	Mod-H↓	Sel.
Phys. Gaussian	3.1M	0.318	0.230	VAL
Honest baseline (UNet1)	17.5M	0.406	0.296	VAL
ConvLSTM ($T=8$)	27.5M	0.603	0.467	ep30
Published RadarHD [1]	17.5M	0.36	0.24	—
<i>Reference (unfair, test-selected):</i>				
Optimized baseline	17.5M	0.295	0.189	TEST
<i>Best on mixed-ID split[†]:</i>				
Phys. Gaussian (exp4)	3.1M	0.280	0.205	VAL
Low-ID subset	—	0.182	0.112	—
High-ID subset	—	0.307	0.250	—

the per-epoch test metrics, showing that the “best” epoch varies significantly and cannot be reliably identified without a proper validation set.

V. EXPERIMENTAL RESULTS

A. Main Comparison

Table II presents the central comparison. All entries use validation-selected checkpoints except where explicitly marked.

The physics-first Gaussian model improves both principal metrics by 22% relative to the honest baseline while using $5.6\times$ fewer parameters. The ConvLSTM model performs significantly worse (Chamfer 0.603 m, more than double the baseline), confirming that late temporal fusion at low spatial resolution is detrimental.

B. Baseline Hyperparameter Sweep

Table III summarizes a systematic hyperparameter sweep of the baseline U-Net on RTX 5090 hardware. The optimal configuration (batch 12, lr 7×10^{-5} , fp32) peaks at epoch 10, with performance degrading at later epochs. This early peaking is consistent across all configurations and suggests that the BCE+Dice loss diverges from geometric metrics after initial convergence.

C. Gaussian Model Loss Tuning

Table IV presents the loss tuning sweep on the Gaussian model. Two key findings emerge: (1) relaxing the sigma prior from 0.1 to 0.3 improves calibration and mod-H by 3%, and (2) adding Huber range loss ($w = 0.1$) reduces range bias, improving mod-H from 0.213 to 0.205 (the best result). Using FPS-based prototypes instead of K -Means significantly hurts performance, as FPS cluster centers provide poorer regression

TABLE III: Baseline U-Net hyperparameter sweep (test-selected, legacy-Cartesian evaluation, epoch in parentheses).

Batch	LR	Prec.	Chamfer↓	mod-H↓
12	7e-5	fp32	0.295 (10)	0.189 (10)
12	7e-5	bf16	0.308 (20)	0.189 (20)
12	1e-4	bf16	0.322 (10)	0.211 (10)
12	5e-5	bf16	0.332 (20)	0.211 (20)
16	1e-4	bf16	0.334 (10)	0.212 (10)
6	1e-4	bf16	0.345 (30)	0.216 (30)

TABLE IV: Gaussian model loss tuning on mixed-ID split ($K=96$, $T=41$, threshold 0.3). Best mod-H in bold.

Config	Proto	mod-H↓	Chamfer↓	Near↓
$\sigma = 0.1$, hub=0	KM	0.219	~ 0.28	0.198
$\sigma = 0.3$, hub=0	KM	0.213	0.281	—
$\sigma = 0.3$, hub=0	FPS	0.324	0.372	—
$\sigma = 0.3$, hub=0.05	KM	0.210	—	—
$\sigma = 0.3$, hub=0.1	KM	0.205	0.280	0.162
$\sigma = 0.3$, hub=0.2	KM	0.225	—	—
$\sigma = 0.5$, hub=0.05	KM	0.215	—	—

K -Means; FPS = Farthest Point Sampling; Near = mod-H at 0–2 m range.

targets. Huber weight above 0.2 overwhelms the NLL term and degrades angular quality.

D. Error Decomposition by Range

Fig. 2 decomposes the Gaussian model’s prediction errors into radial (range) and cross-range (angular) components across distance bins. Near-field accuracy (0–2 m) is strong at 0.227 m total error, with a roughly 60/40 angular/range split. Errors increase monotonically with range, peaking at 6–8 m (0.656 m total), consistent with the Rayleigh resolution limit: angular uncertainty grows with distance for a fixed aperture.

E. Training Dynamics

Fig. 3 shows training curves for both the baseline U-Net and the Gaussian model. The baseline (left) shows monotonically decreasing training loss but non-monotonic validation metrics, confirming that BCE+Dice loss does not directly optimize geometric quality. The Gaussian model (right) shows the NLL component becoming increasingly negative as the model learns tighter, better-calibrated Gaussians, with validation mod-H steadily improving.

F. Oracle Analysis

To understand the headroom in the Gaussian representation itself, we fit K -Means directly to the lidar ground truth. With $K=64$ centroids, the oracle achieves Chamfer 0.031 m and mod-H 0.034 m—an order of magnitude better than any learned model. This confirms that the Gaussian set representation is not the performance bottleneck; the difficulty lies in predicting accurate positions from radar input.

VI. ANALYSIS AND DISCUSSION

A. Why Physics-First Works

The physics-first model’s advantage comes from three complementary factors:

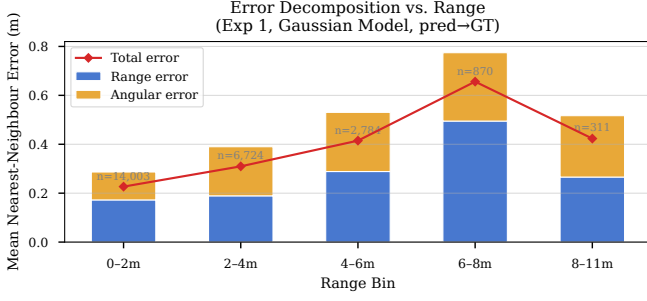


Fig. 2: Prediction error decomposition by range bin. Stacked bars show range (radial) and angular (cross-range) error components. Numbers above bars indicate sample counts per bin.

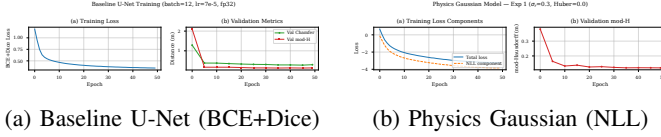


Fig. 3: Training curves. (a) Baseline: training loss decreases but validation Chamfer is non-monotonic. (b) Gaussian: NLL decreases steadily; validation mod-H improves consistently.

Structured input representation. Classical FFT beam-forming provides the encoder with a physically meaningful azimuth–range map rather than a lossy magnitude image. The network only needs to refine this representation, not rediscover signal processing from opaque pixel values.

2D spatial topology preservation. Previous experiments showed that collapsing the azimuth dimension (1D encoders) significantly degrades performance. Sidelobe patterns, multi-target structure, and angular relationships are inherently 2D; preserving this structure through 2D convolutions yields better features for the decoder.

Set-based output with matching supervision. The Hungarian-matched NLL loss encourages one-to-one geometric assignments, directly optimizing point placement precision. In contrast, the baseline’s occupancy loss can produce high F1 scores while placing points imprecisely in space.

B. Why ConvLSTM Failed

The ConvLSTM model’s Chamfer distance of 0.603 m ($2\times$ worse than baseline) is the clearest negative result. The failure is attributable to *where* temporal fusion occurs: the ConvLSTM cells operate at 16×4 and 32×8 spatial resolution after heavy downsampling, by which point fine azimuth–range detail needed for accurate point placement has been destroyed. In contrast, the baseline fuses all 41 frames at the full 256×64 input resolution in the first convolutional layer—a simple but effective strategy for fixed-length sequences where spatial accuracy dominates temporal dynamics.

C. Checkpoint Selection Is a Methodological Concern

The gap between test-selected (0.295 m) and validation-selected (0.406 m) baseline results is 38%. This is not a minor



Fig. 4: Baseline U-Net test metrics across training epochs. The “best” checkpoint varies by epoch and metric, demonstrating why test-set selection produces unreliable comparisons. Horizontal lines show the test-selected optimum vs. honest validation-selected result.

reporting detail—it changes the interpretation of the entire project. Against the test-selected baseline, our model’s improvement would appear marginal; against the honest baseline, it is substantial. Fig. 4 shows that baseline test-set Chamfer fluctuates significantly across epochs with no monotonic trend, making any single epoch choice highly sensitive to luck.

D. Domain Shift Between Trajectory Sets

The mixed-ID split reveals a significant domain gap: on low-ID test trajectories (recorded in similar environments to training data), the model achieves mod-H 0.112 m; on high-ID trajectories (different environments), it degrades to 0.250 m. This $2.2\times$ gap suggests that environment-level generalization, not model capacity, is the current performance bottleneck. Data augmentation (flip, noise, mask) did not measurably reduce this gap.

VII. LIMITATIONS AND IMPROVEMENT PLAN

A. Current Limitations

- 1) **Domain shift.** High-ID trajectories degrade mod-H by $2.2\times$, and augmentation does not help. The root cause is likely environmental differences (scene structure, materials) rather than sensor variation.
- 2) **Far-field accuracy.** Errors at 6–8 m range (0.656 m) are $3\times$ worse than near-field (0.227 m), consistent with the 14° Rayleigh resolution limit for 8 antennas.
- 3) **Validation set reliability.** With only 6–8 trajectories, the validation set is noisy. The val/test gap (e.g., val 0.129 vs. test 0.261 mod-H in one experiment) indicates limited statistical power.
- 4) **Single-frame evaluation.** The Gaussian model’s 41-frame stacking is compared against the baseline’s 41-frame stacking, but a single-frame Gaussian baseline has not been evaluated on the same split for a fully controlled comparison.

B. Planned Improvements

- 1) **Temporal ablation study.** Evaluate performance at $T \in \{1, 3, 5, 8, 41\}$ frames to quantify the value of temporal stacking in the Gaussian model, independent of baseline behavior.

- 2) **Uncertainty calibration analysis.** The model predicts explicit σ_r and $\sigma_\perp$ per Gaussian—we plan to verify whether these correlate with actual spatial error (calibration plots, expected calibration error).
- 3) **Failure-mode decomposition.** Systematic breakdown of errors by scene density, target type (wall, vehicle, vegetation), and angular position.
- 4) **Cross-domain generalization.** Either train on mixed-ID data from the start (already showing promise at mod-H 0.205) or investigate domain adaptation techniques.
- 5) **Inference speed benchmarking.** Measure end-to-end latency on target hardware (Jetson Orin) to validate the <50 ms real-time constraint.

VIII. CONCLUSION

We present initial results demonstrating that a physics-informed radar-to-point-cloud pipeline outperforms image-translation baselines under fair evaluation. The physics-first Gaussian model achieves Chamfer 0.318 m and mod-H 0.230 m—22% improvements over the honestly evaluated baseline—with $5.6\times$ fewer parameters. Equally important, we establish that fair checkpoint selection and explicit dataset splitting are methodological necessities that materially affect reported numbers. The Gaussian set representation with Hungarian matching provides a principled framework for radar point cloud prediction, and error decomposition analysis identifies domain shift and far-field accuracy as the primary remaining challenges.

MEMBER CONTRIBUTIONS

- **Chris Turner:** Baseline U-Net reproduction, hyperparameter sweep, evaluation pipeline development, ConvLSTM implementation and experiments.
- **Andreas Zeck:** Physics-first Gaussian model design and implementation, raw IQ data pipeline, loss function development, loss tuning experiments, fair evaluation protocol, error analysis.
- **Justin Palm:** Dataset preprocessing and management, data split design, training infrastructure (Docker, TensorBoard logging), documentation and report writing.

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